SOFT COMPUTING TECHNIQUES

SCT Practical Viva Questions & Answers

Practical 1 to Practical 9

All questions and answers in practical-file sequence

155 viva questions with simple, exam-ready answers

Includes assignments, sample calculations, key formulas and quick revision

Contents

This guide follows the sequence of the SCT practical file.

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Plus: Common Python Programming Viva Questions

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Study tip: For each practical, learn its aim, inputs, outputs, main algorithm, key formula and the final result shown in your file.

Practical 1: Perceptron Learning Algorithm

Aim: Implementation of the Perceptron Learning Algorithm for Student Admission Prediction.

Q1. What is a Perceptron?

Answer: A Perceptron is the simplest type of Artificial Neural Network. It is used for binary classification and makes decisions based on input values, weights, and bias.

Q2. What is the aim of this practical?

Answer: The aim is to predict whether a student is eligible for admission using entrance examination marks and interview scores.

Q3. What is the Perceptron Learning Algorithm?

Answer: It is a supervised learning algorithm that adjusts weights and bias based on prediction errors to classify data into two categories.

Q4. What are the inputs and outputs of this practical?

Answer: Inputs: entrance examination marks and interview score. Output: 1 means eligible for admission and 0 means not eligible.

Q5. What is supervised learning?

Answer: Supervised learning is a machine learning technique in which a model is trained using labeled input and output data.

Q6. What is a learning rate?

Answer: The learning rate controls how much the weights change during training. In this practical, a learning rate of 0.1 is used.

Q7. What is an epoch?

Answer: An epoch is one complete cycle through the entire training dataset. The Perceptron program specifies a maximum of 50 epochs.

Q8. What are weights and bias?

Answer: Weights determine the importance of each input, while bias helps shift the decision boundary to improve classification.

Q9. What is the Perceptron weight update formula?

Answer: $w_{new} = w_{old} + \eta(y - \hat{y})x$. Here η is the learning rate, y is the actual output, $\hat{y}$ is the predicted output, and x is the input.

Q10. What is a decision boundary?

Answer: A decision boundary is a line that separates data into two classes. In this practical, it separates eligible students from ineligible students.

Q11. Why is normalization used?

Answer: Normalization brings input values into a similar numerical range. In this practical, entrance marks and interview scores are divided by 100.

Q12. What is convergence?

Answer: Convergence occurs when the Perceptron correctly classifies all training samples and the number of classification errors becomes zero.

Q13. Which Python libraries are used?

Answer: NumPy is used for mathematical calculations, and Matplotlib is used to display the decision boundary and learning curve.

Q14. What is the limitation of a single-layer Perceptron?

Answer: It can correctly learn linearly separable datasets, but it cannot solve non-linearly separable problems such as XOR using a single layer.

Assignment 1: Student Placement Prediction

Q15. What is the aim of this assignment?

Answer: To predict whether a student is likely to be placed based on CGPA and coding assessment score.

Q16. What are the input features?

Answer: CGPA and coding assessment score are the two input features.

Q17. What do 0 and 1 represent?

Answer: 0 represents Not Placed, while 1 represents Placed.

Q18. What is the output shown in the practical file?

Answer: For the sample input CGPA 7 and coding score 85, the model predicts that the student is likely to be placed. The displayed training accuracy is 100%.

Assignment 2: Loan Approval Prediction

Q19. What is the aim of loan approval prediction?

Answer: To classify loan applications as Approved or Rejected using annual income and credit score.

Q20. What are the input features?

Answer: The input features are Annual Income and Credit Score.

Q21. What is standardization?

Answer: Standardization transforms values by subtracting the mean and dividing by the standard deviation: $X_scaled = (X - \mu) / \sigma$.

Q22. How is model accuracy calculated?

Answer: Accuracy = (Correct Predictions / Total Predictions) × 100.

Q23. What is the final prediction in the assignment?

Answer: For an annual income of 72,000 and a credit score of 725, the displayed result is Loan Approved.

Practical 2: ADALINE Neural Network

Aim: Implement an Adaptive Linear Neuron using the Delta Learning Rule for House Price Prediction and Loan Approval Prediction.

Q1. What is ADALINE?

Answer: ADALINE stands for Adaptive Linear Neuron. It is a single-layer neural network that uses a linear activation function during learning.

Q2. Who developed ADALINE?

Answer: ADALINE was developed by Bernard Widrow and Ted Hoff.

Q3. What is the aim of this practical?

Answer: To implement ADALINE using the Delta Learning Rule and minimize prediction errors.

Q4. What is the Delta Learning Rule?

Answer: The Delta Rule is a learning method that updates weights using the difference between actual and predicted values to reduce error.

Q5. What is the Widrow-Hoff Learning Rule?

Answer: The Widrow-Hoff Rule, also called the LMS or Delta Rule, adjusts weights to minimize Mean Squared Error.

Q6. What is Mean Squared Error (MSE)?

Answer: MSE measures the average squared difference between actual and predicted values. MSE = $(1/n)\ \Sigma(y_i - \hat{y}_i)^2$.

Q7. What is the main difference between Perceptron and ADALINE?

Answer: A Perceptron uses a threshold-based prediction for learning, whereas ADALINE learns from continuous linear output and minimizes Mean Squared Error.

Q8. Why is data normalization important in ADALINE?

Answer: Normalization makes the training process more stable and helps prevent large input values from dominating weight updates.

Practical 2.1: House Price Prediction

Q9. What are the input parameters for house price prediction?

Answer: The input parameters are area, number of rooms, and location score.

Q10. What is the output of the house price model?

Answer: The output is the predicted house price.

Q11. Which libraries are used in this program?

Answer: NumPy is used for numerical calculations, and Pandas is used to create and manage the dataset.

Q12. What does np.dot() do?

Answer: np.dot() calculates the dot product of arrays, which is used to compute the weighted output of the neural network.

Practical 2.2: Loan Approval Prediction Using ADALINE

Q13. What are the input features?

Answer: Annual income, credit score, years of employment, and existing loan amount.

Q14. What is the output of this program?

Answer: The output is 1 for Loan Approved and 0 for Loan Rejected.

Q15. Why is a threshold of 0.5 used?

Answer: It converts ADALINE's continuous output into a binary class. A value greater than or equal to 0.5 is classified as Approved, otherwise Rejected.

Q16. What accuracy is shown in the practical file?

Answer: The displayed accuracy is approximately 85.71%.

Practical 3: Fuzzy Logic Operators

Aim: Implement fuzzy logic operations for a Smart Traffic Management System.

Q1. What is fuzzy logic?

Answer: Fuzzy logic is a soft computing technique that handles uncertainty by allowing truth values between 0 and 1.

Q2. What is the aim of this practical?

Answer: To use fuzzy AND, OR, and NOT operators to evaluate traffic congestion and road occupancy.

Q3. What is the difference between crisp logic and fuzzy logic?

Answer: Crisp logic uses only two values, 0 and 1. Fuzzy logic allows intermediate membership values such as 0.3, 0.5, and 0.8.

Q4. What is a membership value?

Answer: A membership value represents the degree to which an element belongs to a fuzzy set. It lies between 0 and 1.

Q5. What is the fuzzy AND operator?

Answer: The fuzzy AND operator returns the minimum of two membership values: A AND B = min(A, B).

Q6. What is the fuzzy OR operator?

Answer: The fuzzy OR operator returns the maximum of two membership values: A OR B = max(A, B).

Q7. What is the fuzzy NOT operator?

Answer: The fuzzy NOT operator returns the complement of a membership value: NOT A = 1 − A.

Q8. If traffic congestion is 0.75 and road occupancy is 0.62, calculate fuzzy AND.

Answer: min(0.75, 0.62) = 0.62. Therefore, fuzzy AND = 0.62.

Q9. Calculate fuzzy OR for the same values.

Answer: max(0.75, 0.62) = 0.75. Therefore, fuzzy OR = 0.75.

Q10. Calculate NOT Traffic and NOT Occupancy.

Answer: NOT Traffic = 1 − 0.75 = 0.25; NOT Occupancy = 1 − 0.62 = 0.38.

Q11. Why is fuzzy logic useful in traffic management?

Answer: Fuzzy logic helps manage traffic conditions that are not simply high or low. It supports flexible decisions based on varying congestion and occupancy levels.

Q12. Which graph is used in this practical?

Answer: A bar graph created using Matplotlib displays the fuzzy membership values and the results of the operations.

Practical 4: Classical Sets and Fuzzy Relations

Aim: Perform operations on classical sets, fuzzy sets, fuzzy Cartesian products, and fuzzy relation compositions.

Q1. What is a classical set?

Answer: A classical set is a collection of elements where each element either belongs to the set or does not belong to it. Membership is represented by 0 or 1.

Q2. What is a fuzzy set?

Answer: A fuzzy set allows elements to have different degrees of membership between 0 and 1.

Q3. What is the support of a fuzzy set?

Answer: The support is the set of all elements whose membership values are greater than zero: Support(A) = {x | μ_A(x) > 0}.

Q4. What is the core of a fuzzy set?

Answer: The core contains all elements whose membership values are exactly 1: Core(A) = {x | μ_A(x) = 1}.

Q5. What is the height of a fuzzy set?

Answer: The height is the maximum membership value of a fuzzy set: Height(A) = max μ_A(x).

Q6. What is an alpha-cut?

Answer: An alpha-cut is a classical set containing all elements whose fuzzy membership values are greater than or equal to a specified alpha value: A_α = {x | μ_A(x) ≥ α}.

Q7. What is a normal fuzzy set?

Answer: A fuzzy set is called normal if at least one element has a membership value of 1.

Q8. What is a fuzzy relation?

Answer: A fuzzy relation represents the degree of relationship between elements of two or more sets using membership values between 0 and 1.

Q9. What is the Cartesian product of fuzzy sets?

Answer: The fuzzy Cartesian product combines elements of two fuzzy sets into ordered pairs. In this practical, the minimum membership value is used: μ_(A×B)(x,y) = min(μ_A(x), μ_B(y)).

Q10. What is max-min composition?

Answer: Max-min composition combines two fuzzy relations by first calculating the minimum membership values and then selecting the maximum of those minimum values: t_ij = max_k[min(r_ik, s_kj)].

Q11. What is the difference between a fuzzy set and a fuzzy relation?

Answer: A fuzzy set describes the membership of individual elements, whereas a fuzzy relation describes the degree of association between elements.

Q12. Which mathematical operations are performed in this practical?

Answer: The practical includes calculating the support, core, height, alpha-cut, fuzzy Cartesian product, and max-min composition of fuzzy relations.

Practical 5: Self-Organizing Map (SOM)

Aim: Develop a customer segmentation model using a Self-Organizing Map and retail transaction data.

Q1. What is SOM?

Answer: SOM stands for Self-Organizing Map. It is an unsupervised neural network used for clustering, visualization, and identifying patterns in data.

Q2. Who developed SOM?

Answer: Self-Organizing Maps were developed by Teuvo Kohonen.

Q3. What is the aim of this practical?

Answer: To group customers based on their purchasing behavior using transaction data and a Self-Organizing Map.

Q4. What is unsupervised learning?

Answer: Unsupervised learning is a machine learning technique in which the model identifies patterns or groups from data without predefined target labels.

Q5. What is customer segmentation?

Answer: Customer segmentation is the process of dividing customers into groups based on similar characteristics or purchasing behavior.

Q6. Which dataset is used in this practical?

Answer: The practical uses the online_retail_II.xlsx dataset containing retail transaction records for 2009–2010 and 2010–2011.

Q7. What is RFM analysis?

Answer: RFM stands for Recency, Frequency, and Monetary. Recency means how recently a customer purchased; Frequency means how often the customer purchased; Monetary means how much the customer spent.

Q8. What are the features used for customer segmentation?

Answer: The features are Recency, Frequency, Monetary, Average Order Value, Unique Products, and Total Quantity.

Q9. Why is MinMaxScaler used?

Answer: MinMaxScaler transforms features into a common range, usually 0 to 1, so that features with larger values do not dominate the SOM training.

Q10. What is a Best Matching Unit (BMU)?

Answer: The BMU is the neuron in the SOM whose weight vector is closest to the input data.

Q11. What is a U-Matrix?

Answer: A U-Matrix is a graphical representation of distances between neighboring SOM neurons. It helps identify clusters and their boundaries.

Q12. What is the purpose of K-Means in this practical?

Answer: K-Means is used to group the trained SOM neurons into four clusters, which are then used to classify customers into segments.

Q13. Which Python libraries are used?

Answer: Pandas, NumPy, Matplotlib, Scikit-learn, and MiniSom are used for data preprocessing, visualization, clustering, and SOM training.

Q14. What is the final output of this practical?

Answer: The final output consists of customer groups, a U-Matrix visualization, and a CSV file named Customer_Segments.csv.

Practical 6: Hopfield Neural Network

Aim: Implement a Hopfield Neural Network for binary pattern storage, associative retrieval, energy minimization, and noise tolerance.

Q1. What is a Hopfield Neural Network?

Answer: A Hopfield Neural Network is a recurrent neural network used to store patterns and retrieve them from incomplete or noisy inputs.

Q2. Who developed the Hopfield Network?

Answer: John Hopfield introduced the Hopfield Network model in 1982.

Q3. What is the aim of this practical?

Answer: To store binary patterns, retrieve original patterns from noisy input, and analyze energy minimization and noise tolerance.

Q4. What is associative memory?

Answer: Associative memory is the ability of a network to retrieve a stored pattern when given a partial or noisy version of that pattern.

Q5. Which patterns are stored in this practical?

Answer: The program stores three character patterns: A, B, and C.

Q6. How many neurons are used for the character patterns?

Answer: Each pattern has a 5 × 5 matrix, so 25 neurons are used to represent a pattern.

Q7. What values are used to represent the patterns?

Answer: The patterns use bipolar values, +1 and −1, to represent the binary image pixels.

Q8. What is a recurrent neural network?

Answer: A recurrent neural network contains feedback connections that allow information to influence later network states.

Q9. What is the Hebbian Learning Rule?

Answer: The Hebbian Learning Rule is based on strengthening connections between neurons that activate together. Hopfield networks can use it to construct their weight matrix.

Q10. What is the energy function in a Hopfield Network?

Answer: The energy function measures the stability of the network state: $E = -(1/2) \Sigma_i \Sigma_j w_{ij} s_i s_j$. For the zero-bias network used in this practical, the network attempts to move toward lower-energy stable states.

Q11. What is noise in a Hopfield Network?

Answer: Noise means unwanted changes in the stored pattern. In this practical, noise is introduced by flipping selected pixel values.

Q12. What is noise tolerance?

Answer: Noise tolerance is the ability of the network to recover the correct stored pattern even when some input values have been changed.

Q13. What is convergence in a Hopfield Network?

Answer: Convergence occurs when repeated neuron updates no longer change the network's state.

Q14. What is the purpose of the weight matrix?

Answer: The weight matrix stores the connections between neurons and determines how the network updates and retrieves patterns.

Additional Experiment: Hopfield Network Using MNIST

Q15. What is MNIST?

Answer: MNIST is a dataset of handwritten digit images from 0 to 9. It is commonly used for machine learning experiments.

Q16. What is the aim of the MNIST experiment?

Answer: To store a handwritten digit pattern and study how accurately a Hopfield Network can recover it after introducing noise.

Q17. Why is the MNIST image resized?

Answer: The image is resized to 14 × 14 pixels to reduce the number of neurons and simplify computation.

Q18. How many neurons are used for a 14 × 14 image?

Answer: 14 × 14 = 196 neurons.

Q19. What is the result of the Hopfield experiment?

Answer: The experiment displays the stored, noisy, and retrieved patterns, along with retrieval accuracy, energy history, and noise tolerance results.

Practical 7: Fuzzy Logic-Based Air Conditioner Control

Aim: Design a fuzzy inference system that controls air conditioner fan speed according to temperature and humidity.

Q1. What is the aim of this practical?

Answer: To automatically control AC fan speed using room temperature and humidity with fuzzy logic.

Q2. What is a Fuzzy Inference System (FIS)?

Answer: A Fuzzy Inference System is a decision-making system that uses fuzzy membership functions and IF–THEN rules to produce an output.

Q3. What are the input variables?

Answer: Room temperature and humidity are the input variables.

Q4. What is the output variable?

Answer: Fan speed is the output variable, represented as a percentage from 0 to 100.

Q5. What are the temperature membership categories?

Answer: The temperature categories are Cold, Comfortable, and Hot.

Q6. What are the humidity membership categories?

Answer: The humidity categories are Dry, Normal, and Humid.

Q7. What are the fan speed categories?

Answer: The fan speed categories are Low, Medium, and High.

Q8. What is a membership function?

Answer: A membership function assigns a degree of membership between 0 and 1 to an input value.

Q9. What is fuzzification?

Answer: Fuzzification is the process of converting crisp numerical inputs into fuzzy membership values.

Q10. What are fuzzy IF–THEN rules?

Answer: Fuzzy rules represent conditions and their corresponding decisions. For example, IF temperature is Hot AND humidity is Humid, THEN fan speed is High.

Q11. How many fuzzy rules are used in this practical?

Answer: Nine fuzzy rules are used to handle different combinations of temperature and humidity.

Q12. What is defuzzification?

Answer: Defuzzification is the process of converting fuzzy inference results into a single numerical output.

Q13. What happens when the room temperature is high?

Answer: According to the practical's rules, when the room temperature is Hot, the fan speed is set to High for dry, normal, or humid conditions.

Q14. Which Python library is used to implement the fuzzy system?

Answer: The scikit-fuzzy library is used to define fuzzy variables, membership functions, and rules.

Q15. How is the final fan speed classified?

Answer: The program uses these categories: below 40%: Low; from 40% to below 70%: Medium; 70% and above: High.

Practical 8: Fuzzy Expert System for Medical Diagnosis

Aim: Develop a fuzzy expert system that estimates diabetes risk from blood sugar, BMI, and age.

Q1. What is an expert system?

Answer: An expert system is an AI-based computer program that uses stored knowledge and rules to make decisions similar to a human expert.

Q2. What is a fuzzy expert system?

Answer: A fuzzy expert system combines expert rules with fuzzy logic to make decisions when information is uncertain or has gradual boundaries.

Q3. What is the aim of this practical?

Answer: To estimate diabetes risk using blood sugar, BMI, and age, and compare fuzzy decisions with crisp decisions.

Q4. What are the input features?

Answer: The three input features are Blood Sugar Level, Body Mass Index, and Age.

Q5. What is BMI?

Answer: BMI stands for Body Mass Index. It is a numerical measurement calculated using a person's weight and height: BMI = Weight (kg) / [Height (m)]².

Q6. What is the output of the fuzzy expert system?

Answer: The output is a fuzzy diabetes risk score from 0 to 100, categorized as Low, Medium, or High.

Q7. What are the fuzzy categories for blood sugar?

Answer: The categories used in the practical are Low, Normal, High, and Very High.

Q8. What are the BMI categories?

Answer: The program uses Underweight, Normal, Overweight, and Obese as fuzzy BMI categories.

Q9. What are the age categories?

Answer: The fuzzy age categories are Young, Middle, and Old.

Q10. What is a rule base?

Answer: A rule base is a collection of IF–THEN rules used by the expert system to make decisions based on different input conditions.

Q11. What is the difference between crisp and fuzzy decisions?

Answer: A crisp decision uses fixed thresholds, whereas a fuzzy decision uses membership functions and multiple rules to consider gradual changes.

Q12. How is the final risk score classified?

Answer: According to the program: below 40 is Low risk; from 40 to below 70 is Medium risk; 70 and above is High risk.

Q13. What sample input is shown in the practical file?

Answer: The sample input consists of a blood sugar level of 96 mg/dL, a BMI of 35, and an age of 20 years.

Q14. What is the sample output?

Answer: The fuzzy system gives a risk score of 50% and classifies it as Medium, while the crisp comparison classifies it as Low.

Q15. Why are the crisp and fuzzy results different?

Answer: The crisp example uses a fixed blood sugar threshold, whereas the fuzzy system considers blood sugar, BMI, and age together.

Q16. What is the advantage of a fuzzy expert system?

Answer: It can handle uncertainty and gradual differences in input conditions, making it useful for decision-support experiments.

Exam note

This practical is an academic demonstration. Its output is not a clinically validated diabetes diagnosis or a medical risk probability.

Practical 9: Hybrid Neural Network and Fuzzy Logic

Aim: Develop a hybrid soft computing system that classifies waste as Dry or Wet using a neural network and fuzzy logic.

Q1. What is a hybrid intelligent system?

Answer: A hybrid intelligent system combines two or more intelligent computing techniques to perform a task. In this practical, it combines a neural network and fuzzy logic.

Q2. What is the aim of this practical?

Answer: To classify waste as Dry or Wet using moisture, weight, and texture information.

Q3. What are the input features?

Answer: The input features are Moisture Level, Weight, and Texture Score.

Q4. What are the output classes?

Answer: There are two output classes: 0 represents Dry Waste, and 1 represents Wet Waste.

Q5. Which neural network is used?

Answer: The practical uses an MLPClassifier, which is a Multi-Layer Perceptron neural network.

Q6. What is an MLP?

Answer: A Multi-Layer Perceptron is a feedforward artificial neural network that contains an input layer, one or more hidden layers, and an output layer.

Q7. What is the activation function used in this practical?

Answer: The ReLU activation function is used. ReLU stands for Rectified Linear Unit: ReLU(x) = max(0, x).

Q8. What is the purpose of StandardScaler?

Answer: StandardScaler standardizes the input features so they have comparable scales during neural network training.

Q9. What is the train-test split?

Answer: Train-test split divides the dataset into separate training and testing portions. In this practical, 25% of the data is reserved for testing.

Q10. How is fuzzy logic used in waste classification?

Answer: Fuzzy logic calculates wet and dry membership values using moisture and texture, then applies rules to estimate whether the waste is Wet or Dry.

Q11. How are the neural network and fuzzy logic combined?

Answer: The final probability is calculated using a weighted combination: P_hybrid = 0.60 × P_NN + 0.40 × P_fuzzy. The Neural Network contributes 60%, and Fuzzy Logic contributes 40%.

Q12. What is the classification threshold?

Answer: A threshold of 0.50 is used. If the hybrid wet probability is greater than or equal to 0.50, the waste is classified as Wet; otherwise, Dry.

Q13. What is a confusion matrix?

Answer: A confusion matrix is a table used to compare actual and predicted classes. It helps identify correct and incorrect classifications.

Q14. What is precision?

Answer: Precision measures how many of the samples predicted as a particular class actually belong to that class.

Q15. What is recall?

Answer: Recall measures how many of the actual samples belonging to a class are correctly identified by the model.

Q16. What is an F1-score?

Answer: F1-score is the harmonic mean of precision and recall. It provides a combined measure of classification performance.

Q17. What accuracy is shown in the practical file?

Answer: The hybrid model achieves 100% accuracy on the eight test samples shown in the file. This does not guarantee the same accuracy on new datasets.

Q18. What is the final prediction for the sample waste item?

Answer: For moisture 72%, weight 450 g, and texture score 3.5, the model predicts Wet Waste, with a displayed hybrid wet probability of approximately 0.984.

Common Python Programming Viva Questions

Q1. Why is Python used for Soft Computing?

Answer: Python is easy to learn and provides useful libraries for neural networks, machine learning, numerical calculations, and visualization.

Q2. What is NumPy?

Answer: NumPy is a Python library used for arrays, matrices, and mathematical calculations.

Q3. What is Pandas?

Answer: Pandas is a Python library used for data manipulation and analysis. It provides structures such as DataFrames.

Q4. What is Matplotlib?

Answer: Matplotlib is a Python library used to create graphs, charts, and data visualizations.

Q5. What is Scikit-learn?

Answer: Scikit-learn is a machine learning library that provides algorithms and tools for classification, clustering, scaling, and model evaluation.

Q6. What is the difference between classification and regression?

Answer: Classification predicts categories such as Approved or Rejected. Regression predicts numerical values such as house prices.

Q7. What is overfitting?

Answer: Overfitting occurs when a model learns the training data too closely, including noise, and performs poorly on new data.

Q8. What is an activation function?

Answer: An activation function processes a neuron’s weighted input and determines its output.

Q9. What is the difference between supervised and unsupervised learning?

Answer: Supervised learning uses labeled data, whereas unsupervised learning finds patterns or groups without target labels.

Q10. What is the main purpose of Soft Computing?

Answer: Soft Computing aims to solve complex real-world problems by handling uncertainty, approximation, incomplete information, and adaptive learning.

Quick Revision Table

Practical	Main topic	What to remember
1	Perceptron	Binary classification, weights, bias, epochs
2	ADALINE	Delta Rule, MSE, linear output
3	Fuzzy Operators	AND = Min, OR = Max, NOT = 1 − x
4	Fuzzy Sets	Support, core, height, alpha-cut, max-min
5	SOM	Unsupervised learning, clustering, U-Matrix
6	Hopfield Network	Associative memory, energy, noise tolerance
7	Fuzzy AC Control	Temperature, humidity, nine rules, fan speed
8	Fuzzy Expert System	Blood sugar, BMI, age, crisp vs fuzzy
9	Hybrid System	MLP, fuzzy logic, 60:40 combination

Exam preparation tip: Learn the aim, algorithm, inputs, outputs, important formula and final result of every practical. These points help you explain your program clearly to the examiner.